

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and configured **Control Modules**, **Equipment Modules**, and **Phase Classes** within the **Emerson DeltaV DCS** platform for a regulated manufacturing environment.
- Collaborated with Process Subject Matter Experts and CQV Leads to gather and translate process requirements into functional design specifications (FDS) for automation systems.
- Implemented code changes and system configurations within the DeltaV environment, ensuring adherence to project specifications and client needs.
- Participated in and signed off on key functional design aspects, contributing to the overall architecture of automation solutions for life science customers.
- Supported internal project reviews with stakeholders and clients, presenting technical designs and implementation progress.
- Managed the automation compliance lifecycle and change management processes, ensuring adherence to industry regulations and computer system validation standards.
- Containerized automation scripts and deployment tools using **Docker** for consistent execution across development and testing environments on **AWS EC2**.

SKILLS

Automation Systems – Emerson DeltaV, DCS, PLC, SCADA, Machine Vision, ISA-88, GAMP, GMP

Programming & Configuration – DeltaV Coding, Control Modules, Equipment Modules, Phase Classes, Python, Java

Cloud & DevOps – AWS, EC2, Docker, Jenkins, Git

Data & Analytics – Power BI, Tableau, DAX, Power Query, pandas, NumPy

Databases – PostgreSQL, MySQL, Oracle, SQL

Methodologies – Agile, Scrum, Change Management, Computer System Validation, Process Control

PROJECTS

DeltaV Automation System Design for Life Sciences

- Designed and configured **Emerson DeltaV Control Modules** and **Equipment Modules** for a simulated biopharmaceutical manufacturing process, focusing on ISA-88 batch control principles.
- Developed Python scripts to automate the generation of test cases for DeltaV configurations, reducing manual testing effort by approximately 30%.
- Documented system designs and validation protocols in accordance with GAMP and GMP regulatory requirements for a regulated manufacturing environment.

Production Support Automation with Python and AWS

- Automated routine production support tasks for enterprise applications using **Python scripting**, including log analysis and data extraction from **Splunk**, reducing manual triage time by 2 hours per incident.
- Developed **AWS Lambda** functions triggered by CloudWatch events to monitor system health and generate alerts, improving incident detection speed by 15%.
- Wrote **SQL queries** against PostgreSQL databases to validate data integrity and support reconciliation workflows for financial transaction systems.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)